MIL-P-46464A (MU)
4 October 1967
SUPERSEDING
MIL-P-46464 (Ord)
30 December 1960

MILITARY SPECIFICATION

PELLETS, TETRYL

1. SCOPE

1.1 Scope.-This specification covers two types of tetryl pellets for use in ammunition.

1.2 Classification.-Tetryl pellets shall be of the following types as specified (see 6.1).

Type I - Pellets made from tetryl alone or from tetryl and graphite.
Type II - Pellets made from tetryl with a binder-lubricant. The binder-lubricant may consist of graphite, barium stearate, and calcium stearate or graphite and barium stearate, or graphite and calcium stearate, or calcium stearate, or barium stearate or calcium stearate and barium stearate.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-T-45 - Tape, Gummed Paper Reinforced and Plain For Sealing and Securing.
PPP-F-320 - Fiberboard, Corrugated and Solid Sheet Stock Container, Grade and Cut Shapes.
PPP-B-636 - Box, Fiberboard

MILITARY

MIL-G-155 - Graphite (For Use in Ammunition)
MIL-C-263 - Calcium Stearate
MIL-T-339 - Tetryl (Trinitrophenylmethylnitramine)

FSC: 1376

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MIL-B-366 - Barium Stearate
MIL-B-2427 - Boxes, Ammunition Packing: Wood, Nailed.
MIL-L-10547 - Liners, Case Waterproof
MIL-B-13239 - Barrier Material, Water-proof, All Temperatures, Flexible.
MIL-A-13374 - Adhesive, Dextrin (Spiral Tube Winding for Ammunition Containers).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-650 - Explosive, Sampling, Inspection and Testing
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

INTERSTATE COMMERCE COMMISSION

49 CFR 71-90 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1947 Edition and revisions) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. D. Orders fro the above publications should cite "49 CFR 71-90 (latest revision).)"

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3. REQUIREMENTS

3.1 Material.-The constituent materials used in the manufacture of tetryl pellets shall comply with the following requirements:

Constituent Material	Conforming to:
Tetryl	MIL-T-339, Grade I Class a or b
Graphite	MIL-G-155, Grade I or II
Barium Stearate	MIL-B-366
Calcium Stearate	MIL-C-263

3.2 Moisture.-The moisture content of the unconsolidated (loose state just prior to pelleting) tetryl, tetryl and graphite and tetryl and binder-lubricant shall not exceed 0.10 percent when determined as specified in 4.3.1.

3.3 Composition.-The analytical composition of the tetryl pellets shall be as follows, when determined as specified in the applicable paragraph of 4.3.

Pellets	Constituent, percent		
	Tetryl	Graphite	Binder-Lubricant
Type I	98.0 minimum (min)	2.0 maximum (max)	--
Type II (see 6.3)	98.0 min.	--	2.0 max.

3.4 Cracks or laminations.-The tetryl pellets shall be free of cracks and laminations (see 4.2.2.1).

3.5 Detail requirements.-The detail requirements pertaining to density, dimensions and weight of the tetryl pellets shall be as specified on the applicable drawing.

3.6 Workmanship.-All pellets and packaging material shall be free of dirt, grease and other foreign matter.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein.

4.1.1 Contractor quality assurance system.-If the contractor desires to utilize a quality assurance system, which is at variance with the quality assurance provisions of 4.2 and 4.3 and other documents referenced herein, he shall submit a written description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection (measuring and testing equipment) and provisions for control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance provisions required by the provisions of 4.2 and 4.3 and the other documents referenced herein. The contractor shall not be restricted to the inspection station or the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any changes to the written procedure that effects the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Quantity of product inspected.

c. Results obtained for all inspection performed.

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d. Specification number and date, together with an identification and date of changes.

e. Certificates of analysis on all material procured directly by the contractor, when such material is covered by referenced Government specification.

f. Quantity of product in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure (see 4.1.1), this detail specification and other contractual documents as a guide, the Government inspector shall verify all quality assurance operations performed by the contractor. Verification shall be in accordance with a or b as applicable, the decision being the responsibility of the procuring activity. In either case, the inspector shall also ascertain, prior to acceptance, that all quality assurance provisions of other specifications referenced in any of the contractual documents have been complied with. Deviations from prescribed or agreed upon procedures discovered by the Government Inspector shall be brought to the attention of the supplier. Disposition of the product and remedial action shall be as directed by the Government inspector and depending on the nature of the deviation, may consist of lot rejection, screening, re-sampling, re-instruction of the supplier's employees, or other appropriate action.

a. Verification at the point of manufacture shall be accomplished at unscheduled intervals in accordance with 4.1.3.1 and 4.1.3.2.

b. Verification at the point of delivery shall be in accordance with 4.1.3.2.

4.1.3.1 Surveillance.-Surveillance shall include, but is not limited to:

a. Observation of procedures concerning lot formation and identification.

b. Observation of sampling procedures and application of acceptance criteria.

c. Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or of approved equivalents thereto.

d. Review of procedures for control and disposition of non-conforming material.

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4.1.3.2 Product inspection.-Product inspection shall consist of Government inspection of product which has been previously inspected by the contractor and found to meet the quality assurance provisions of this specification. The inspection by the Government shall be performed in order to determine that the product is of the quality required by this specification and that the contractor's records are reliable.

4.2 Inspection provisions

4.2.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and insepcted to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, specification or revision thereof. Changes to either the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government inspector, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105 and shall be numbered in accordance with MIL-STD-1168. Unless otherwise approved by the contracting officer, the inspection lot size of major assemblies or end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period. Inspection lots for components or subassemblies other than the items of delivery, shall be homogeneous and of a size convenient to the contractor.

4.2.2 Examination.-Inspection for critical defects (and Major defects, when so specified) shall be 100 percent.

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Sampling plans and procedures for Major and Minor defects shall be in accordance with Standard MIL-STD-105 except that continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual ~~characteristics listed using an AQL of 0.65~~ percent for each Minor defect and an AQL of 0.40 percent for each Major defect, except where 100 percent inspection is specified.

4.2.2.1 Tetryl pellets (see applicable drawing)

Categories	Defects	Method of Inspection	Code No. (see 6.5)
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Weight (when applicable)	Balance	01001
102.	Outside diameter	Gage	01002
103.	Inside diameter (when applicable)	Gage	01003
104.	Length	Gage	01004
105.	Crack or lamination (see 3.4)	Visual	01005
106.	Evidence of poor workmanship (see 3.6)	Visual	01006
Minor:	None defined.		

4.2.2.2 Carton, sealed

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Contents loose	Manual	02001
102.	Stapling or stitching improper or insecure (see 5.1.1)	Visual	02002
103.	Weight of explosive, max.	Balance	02003

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Minor:	AQL 0.65 percent		
201.	Marking misleading or unidentifiable	Visual	02004
202.	Carton closure insecure (see 5.1.1)	Visual-Manual	02005
203.	Evidence of poor workmanship (see 3.6)	Visual	02006

4.2.2.3 Box, wood packing, sealed

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Box sprung or board broken or split to extent that contents are exposed	Visual	03001
102.	Strapping missing, broken or loose	Visual-Manual	03002
Minor:	AQL 0.65 percent		
201.	Contents loose	Manual	03003
202.	Strapping improperly assembled	Visual	03004
203.	Marking misleading or unidentifiable	Visual	03005

4.2.3 Testing

4.2.3.1 Composition (see applicable drawing.)-Five pellets shall be removed from the beginning of the pelleting operation and five pellets shall be removed from the end of the pelleting operation. The two five pellets samples shall be crushed and packed as two separate samples. The samples shall be tested for the composition requirements. If any sample fails to meet the requirement the batch of pellets shall be rejected.

4.2.3.2 Density (when applicable).-The sampling plan shall be in accordance with Table I.

TABLE I

Sample	Sample size	Cumulative Sample size	Acceptance Number	Rejection Number
First	10	10	0	2
Retest	10	20	1	2

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4.3 Test methods and procedures

4.3.1 Moisture content, Major defect, Code No. 04001.- An accurately weighed portion of approximately 5 grams (gm) of the sample shall be transferred to a tared weighing dish with a cover. The cover, and the uncovered dish and contents shall be placed in an oven maintained at 100 ± 5 degrees Centigrade (^{o}C) for 1 hour. The dish shall be covered and placed in a desiccator to cool. Weigh and calculate the percent moisture as follows:

$$\text{percent moisture} = \frac{100\ A}{W}$$

where:

A = loss in weight of dish and contents.
W = weight of sample.

4.3.2 Determination of total graphite content (applicable to Type I only) or Determination of total binder-lubricant (applicable to Type II only) (see 6.3 and 6.4), Major defect, Code No. 05001.-An accurately weighed portion of 5 gm. of the sample shall be transferred to a tared, medium porosity, sintered glass filtering crucible. The sample shall be extracted with four 10-milliliter (ml) portions of acetone, each portion shall remain in contact with the residue for approximately five minutes before applying suction. Any lumps which may persist shall be broken up and the sides of the crucible shall be washed down with each 10 ml. portion of acetone to insure complete removal of the tetryl. The residue shall be aspirated until the odor of acetone is no longer detectable. The crucible and its contents shall be placed in an oven maintained at 100° + 5°C. for ½ hour, cooled to room temperature in a desiccator and then weighed. The residue shall then be re-extracted with 10 ml. portions of acetone, repeating the suction, drying, cooling and weighing operations after each additional portion, until two successive weighings do not differ from each other by more than 0.002 gm. Calculate the total percentage of binder-lubricant as follows:

$$\text{Total percent of binder-lubricant or total graphite} = \frac{100\ A}{B}$$

where:

A = weight of residue in crucible.
B = weight of specimen on a moisture-free basis.

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4.3.3 Tetryl content, Major defect, Code No. 06001.-The percent tetryl content shall be determined by subtracting from 100 percent the percent total binder-lubricant or total graphite (as applicable) obtained as specified in 4.3.2.

4.3.4 Density, Major defect, Code No. 07001.-The density shall be determined in accordance with Method 203.1 of Standard MIL-STD-650.

5. PREPARATION FOR DELIVERY

5.1 Packaging

5.1.1 Level A and Level B - Unit Pack.-Unless otherwise specified, the pellets shall be tightly packaged in convolute or spiral wound chipboard tubes of the proper inside diameter to accommodate the pellets and not exceeding 15 inches in length. The chipboard tubes shall be four ply of 0.16 tube winding stock with dextrin adhesive, Specification MIL-A-13374, applied between layers. When spirally wound, all layers shall have butt joints with a maximum gap of 1/16 inch allowed. The inside of the tubes shall be closed with end caps of 0.25 inch cover stock. The tubes shall be sealed with one end cap glued to the body tube and the opposite end cap sealed with one layer of adhesive tape, around the joint of the cap and the body tube. The tape shall comply with requirements of Type III, Grade B of Specification PPP-T-45. The pellets shall be cushioned at each end of the tubes with 3/16 inch thick pressed felt cut to the proper inside diameter of the tubes.

5.1.2 Level C - Unit Pack.-RDX pellets shall be unit packaged as specified in 5.1.1.

5.1.3 Level A and Level B - Intermediate Pack.-Tubes containing the pellets shall be packed horizontally in a fiberboard carton complying with requirements of Specification PPP-B-636, Style RSC, Grade 175, B Flute, Class Domestic cartons. The tubes shall be separated from the carton and between layers by a minimum of one inch of clean dry and tightly packed sawdust. Not more than 10 pounds, net weight, of explosive shall be packed in one carton. The carton shall be closed with two inch wide tape conforming to Type III, Grade B of Specification PPP-T-45. Each carton shall be labeled to describe the contents.

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5.1.4 Level C - Intermediate Pack.-Tubes containing the pellets shall be packed as specified in 5.1.3.

5.2 Packing

5.2.1 Level A and Level B.-Five (5) intermediate packages shall be packed in a Grade A, Type I, Class 2, wood box conforming to Specification MIL-B-2427. Case liners fabricated in accordance with Specification MIL-L-10547, Type I, Grade A using Type CW-2 Grade A material conforming to Specification MIL-B-13239 shall be used in each box. The length and width of the liner shall be approximately ¼ inch greater than the inside length and width of the box. After sealing the case liner, there shall be sufficient looseness between the contents and the liner to insure no pressure being applied to liner or seams. Both ends and the top of the box shall be lined with W5S or W6S fiberboard conforming to Specification PPP-F-320. The box shall be closed and strapped in accordance with the Appendix of the box specification.

5.2.2 Level C.-The pellets shall be packed as specified in 5.2.1 except box shall be Class I, Grade C, and fiberboard filler shall be class domestic, Grade 175.

6. NOTES

6.1 Intended use.-Tetryl pellets are intended for the following uses:

Type I - for use in detonators, leads and boosters
Type II - for use in boosters only

6.2 Ordering data.-Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Type required (see 1.2).
c. Level of protection (see 5.1 and 5.2).

6.3 Binder lubricant.-It is advisable that the binder-lubricant used for Type II tetryl pellet contain at least 0.25 percent graphite.

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6.4 Type II.-The following procedures, applicable to Type II pellets only, are optional and are included in this Specification for use in the event that the determinations are desired.

6.4.1 Graphite content.-Fill the crucible containing the residue from the tetryl extraction, obtained as specified in 4.3.2 approximately 2/3 full of 3 normal (N) hydrochloric acid and place in a 250 ml. beaker. Add sufficient 3 N hydrochloric to the beaker to cover approximately ¼ of the crucible, and cover the beaker with a watch glass. Place the beaker and contents on a hot plate; bring the solution to a boil and allow to boil gently for 30 minutes. Filter the mixture in the crucible, using suction, and wash the residue with five 20-ml. portions of hot distilled water followed by three 20 ml. portions of acetone. Aspirate the crucible and contents to remove acetone, dry in an oven at 100 to 105°C. for one-half hour, cool in a desiccator and weigh. Calculate the weight of the residue in the crucible to the percentage of graphite in the total binder-lubricant as follows:

$$\text{Percent graphite} = \frac{100\ A}{W}$$

where:
A = increase in weight of empty, tared crucible.
W = weight of sample on a moisture-free basis.

6.4.2 Stearate content (as barium and/or calcium stearate.-Calcium the percent stearate in the binder-lubricant by subtracting the percent graphite found, as specified in 6.4.1 from the percent total binder-lubricant obtained, as specified in 4.3.2.

6.5 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

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Army-MU

Preparing activity:
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